

K Shiv Kumar (Shiv)

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Software/Systems Engineer with a strong foundation in networking and low-level systems, combining **1.8 years** of experience in industry with **more than 4 years** of research in programmable data planes (P4 and eBPF) and compiler framework (LLVM). Experienced in C/C++, Python, Linux Networking Stack, Host Networking, Linux Operating System, and Systems programming, with a proven ability to build solutions for real-world systems challenges.

EDUCATION

IIT Hyderabad - (NETX Lab) Kandi, Telangana
MS (Computer Science and Engineering) - (GPA: 7.60) 2021–2026

Research: eBPF, P4 and Compilers

IIIT Naya Raipur Naya Raipur, Chhattisgarh
MTech (Computer Science and Engineering) - (CGPA: 8.78/10) 2019–2021

Shri Shankaracharya Engineering College Bhilai, Chhattisgarh
B.E. (Information Technology and Engineering) - (CGPA: 8.17/10) 2011–2015

TECHNICAL SKILLS

Core: eBPF(XDP/TC), P4, C++, Python

Libraries and Tools: Wireshark, Mininet, Docker, Kubernetes, Github, SQL, Libbpf

Frameworks: BMv2, Tofino, Barefoot Runtime Interface (BRI), LLVM

Networking Concepts: Network fundamentals, Linux networking stack, Programmable networks, SDN, AWS

Miscellaneous: Technical Documentation, Linux shell scripting, Virtual Machines

PROJECTS

“Yaksha-Prashna: Understanding eBPF Bytecode Network Function Behaviour” | ACM CoNEXT '26: (🔗)

- Developed a system to validate the correctness and interactions of third-party eBPF network function bytecode.
- Applied program analysis to extract functional properties of eBPF network functions from the bytecode.
- **Tools & Frameworks:** C++, LLVM, eBPF/XDP, Libbpf, Linux kernel, bpftool, bpfman.

“In-Network Probabilistic Monitoring Primitives under the Influence of Adversarial Network Inputs” | APNET '23: (🔗)

- Analyzed vulnerabilities of monitoring primitives in programmable data planes against adversarial inputs.
- Demonstrated FlowRadar pollution attacks where a few malicious flows corrupted telemetry data.
- Quantified attack impact, revealing a 99% drop in FlowRadar’s accuracy under targeted adversarial conditions.
- **Tools & Frameworks:** Python, Wireshark, MS Excel.

“DBVal: Validating P4 Data Plane Runtime Behaviour” | ACM SOSR '21: (🔗)

- Developed DBVal, a runtime validation system to detect packet-processing errors in P4 data planes.
- Implemented assertion-based tracking of tables and actions, enabling line-rate validation of runtime Behaviour.
- **Tools & Frameworks:** Python, P4 language, Lex & Yacc, BMv2, Mininet, Wireshark, VirtualBox.

PROFESSIONAL EXPERIENCE

Application Development Analyst (Accenture, Pune, India) Mar 2016–Dec 2017

- Integrated a Learning Management System and developed an off-boarding feature for the client’s Salesforce environment, enhancing user onboarding and lifecycle workflows.
- **Award:** Apex Award - Extra Miller, in Accenture India.

CERTIFICATIONS

- **Intel Connectivity Academy Level 1** | Issued by: Intel (2021). (🔗)
 - Covered data/control plane development with training in advanced parsing, externs in TNA (Tofino Native Architecture), Barefoot Runtime Interface, and Tofino ASIC architecture.

AWARDS AND HONORS

- ACM SIGMETRICS Student grant: For presenting our ongoing research on “*Detecting Adversarial Attacks on Bloom Filters in P4 Data Plane Systems*” (2022). ([🔗](#))
- COMSNETS Travel Grant: For attending the Conference (2022). ([🔗](#))

TALKS AND PRESENTATIONS

MS, IIT Hyderabad

2021–2025

- Attended first eBPF Day India workshop organized by IISc and eBPF Foundation (2024). ([🔗](#))
- Conducted a session titled “*Introduction to eBPF*” in a SERB-sponsored Workshop on “SDN: Software Defined Networking Architectures and Applications” at IIIT Kottayam (2023). ([🔗](#))
- Delivered a seminar talk on “*Introduction to eBPF*” in CSE Department seminar series at IIT Hyderabad (2023)
- Presented the work “*Detecting Adversarial Attacks on Bloom Filters in P4 Data Plane Systems*” at the Graduate Forum, PerfNA Workshop, IIT Bombay (2022). ([🔗](#))

VOLUNTEERING AND ASSISTANTSHIPS

MS, IIT Hyderabad

2021–2025

- Teaching Assistant for courses *Computer Networks*, *Advanced Computer Networks* and *Topics in Networks*.
- Volunteered in the *International Conference on Distributed Computing and Networking*. ([🔗](#))
- Volunteered in *ACM Workshop on Research Opportunities in Computer Science*. ([🔗](#))

MTech, IIIT Naya Raipur

2020–2021

- Teaching Assistant for courses *Advanced Computer Networks* and *Software Defined Networks*.